Startup Investment Ecosystem

Exploratory Data Analysis Report

Prepared for: University Machine Learning Project

Evidence base: Existing notebook analysis and saved outputs

Data provenance: Not identified in the notebook

A detailed, evidence-based account of the existing exploratory analysis. No original notebook cells or source data were changed.

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Prepared for: University Machine Learning Project **Analysis source:** Existing Startup_Investments.ipynb notebook and its saved outputs **Dataset source/provenance:** Not identified in the notebook **Report status:** Documentation of completed EDA; no model was trained

> **Evidence note.** This report documents saved notebook results and visualizations. It does not rerun the EDA, change source data, or modify the notebook. Interpretations based only on field names are labeled as inferred. No cleaning, imputation, outlier treatment, or target selection is presented as completed work.

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1. Executive Summary

The notebook examines a startup and investment ecosystem represented by 11 related CSV tables. Together, the tables contain **1,500,237 rows and 154 columns**. The largest table, objects.csv, contains 462,651 records and 40 fields. The table names and fields indicate coverage of companies and other entities, people, funding rounds, investments, offices, degrees, funds, milestones, acquisitions, IPOs, and relationships. This subject-matter description is inferred from the schema; the dataset's publisher, collection period, and formal data dictionary are not documented.

The EDA found **9,260,295 missing cells** across the tables, with substantial variation by table and field. No exact duplicate rows were reported in any table. The notebook's inferred candidate primary-key checks found no missing or duplicate candidate keys. Date-like strings were sample-parseable at 100% for the non-missing sampled values in the checked fields, but were not converted in the source tables. Potential outliers were flagged by IQR and Z-score rules; they were **not removed, capped, or transformed**.

In the 196,553 records filtered as companies, the candidate status label is dominated by operating (183,441; 93.33%). The remaining observed categories are acquired (4.78%), closed (1.31%), and IPO (0.58%). This is a highly imbalanced outcome if treated as a classification target, and it is not a confirmed prediction objective.

The most consequential statistical pattern is a long right tail in financial and count variables. For example, funding-round raised_amount_usd has a median of $1.6 million and a maximum of $3.835 billion; its skewness is 50.22. This gap between typical and extreme values means means and standard deviations can be strongly influenced by a small number of very large observations.

Overall ML readiness: exploratory, not training-ready. The dataset is rich and has checked candidate keys, but it is a set of linked tables rather than a ready-made company-level modeling matrix. A specific prediction question, target, prediction date, correct relational joins, domain review of missing values/zeros/outliers, and leakage-safe preprocessing remain necessary.

2. Dataset Overview

2.1 Scope and dimensions

The notebook loaded 11 CSV files from the project's local data/ folder. The table rows should not be added together and treated as unique companies: the tables have different record grains (for example, one record per office, funding round, or relationship), and entities can appear in multiple tables.

Table	Rows	Columns	Missing cells	Missing cells (%)	Exact duplicate rows
objects.csv	462,651	40	7,917,475	42.78	0
relationships.csv	402,878	11	511,067	11.53	0
people.csv	226,709	6	198,663	14.60	0
offices.csv	112,718	15	431,467	25.52	0
degrees.csv	109,610	8	91,144	10.39	0
investments.csv	80,902	6	0	0.00	0
funding_rounds.csv	52,928	23	78,511	6.45	0
milestones.csv	39,456	9	18,243	5.14	0
acquisitions.csv	9,562	12	9,658	8.42	0
funds.csv	1,564	11	753	4.38	0
ipos.csv	1,259	13	3,314	20.25	0
Total	**1,500,237**	**154**	**9,260,295**	—	**0**

The missing-cell percentage is reported per table as in the notebook. A single overall percentage is not reported here because tables have different row grains, and combining all cells would not by itself describe the usable coverage of any modeling population.

2.2 Tables and their roles

Table	Plain-language description inferred from table and field names
objects.csv	Registry of entities such as companies, people, and products, with names, locations, status, and summary counts.
people.csv	Person records and basic identity/affiliation information.
degrees.csv	Education or degree records linked to objects/people.
offices.csv	Organization office descriptions, locations, and coordinates.
funding_rounds.csv	Funding event details, round types, amounts, valuations, and participant counts.
investments.csv	Links between funding rounds, funded entities, and investor entities.
funds.csv	Fund records, including reported name, funding date, and amount.
milestones.csv	Dated milestone records associated with entities.
acquisitions.csv	Acquisition-event parties, date, terms, and price fields.
ipos.csv	IPO/public-listing event, valuation and raised amount, currency, and ticker fields.
relationships.csv	Person-to-entity relationship records, including role/title and date fields.

The notebook does not provide an external data dictionary. These descriptions are best-effort interpretations of the field names and should be checked against the original dataset documentation before making domain-sensitive decisions.

2.3 Feature types and possible target

The notebook's numerical summary includes **22 numeric features** across the tables, while its categorical frequency summary covers **19 categorical fields**. Other columns are inferred as identifiers, date-like fields, free text/high-cardinality strings, or binary indicators. These counts reflect the notebook's role-classification and reporting logic; they are not a claim that every other field is a useful model feature.

objects.csv contains 40 columns. The notebook identified company records using entity_type == "Company" and reported **196,553 company rows**. status could be investigated as a candidate classification label, but the notebook explicitly did not select it as a target. A prediction task needs a defined outcome and the date at which a prediction is meant to be made.

2.4 Potential real-world questions

Subject to representative coverage, correct joins, and a clear time boundary, the data could help investigate:

- How funding-round amounts differ by round type, time, or company attributes.
- How recorded company lifecycle statuses vary across selected categories.
- How investment and relationship networks connect organizations and investors.
- How acquisitions and IPO records are distributed among entities.

- Whether company-level attributes available at a historical cutoff are associated with later outcomes.

These are possible analyses, not findings of causation or a confirmed project objective.

3. Understanding the Problem

The notebook is an exploratory analysis of a multi-table business dataset, not a completed prediction study. It can describe observed records and associations, but the available notebook does not establish who was eligible to be included, how records were sampled, when the dataset was collected, or what "success" means.

Questions it can support

- What entity and event tables are available, and how much information is missing?
- What are typical and extreme reported funding, acquisition, fund, and IPO amounts?
- How are numeric measurements distributed, and which observations trigger conventional outlier screens?
- How do the recorded statuses vary across the ten most common company categories shown in the notebook?
- Which numeric variables have the strongest within-table linear correlations?

Task types and inference boundaries

- **Classification:** Possible only after a target is approved. Company status is a candidate, not a confirmed label.
- **Regression:** Possible for an amount or valuation field only after defining the prediction population, time point, currency convention, and handling of zero/missing values.
- **Clustering or network analysis:** Plausible with carefully selected entity attributes or relationship/investment edges, but not performed in the notebook.
- **Descriptive analysis:** Supported by the saved tables and plots.

The notebook does not prove that funding causes a company to succeed, that a recorded status is a future outcome, or that the observed records represent all startups. Correlation and grouped percentages are descriptive associations.

4. Data Quality Assessment

4.1 Missingness

The notebook found missing values in **68 of 154 columns**. Missingness ranges from no missing cells in investments.csv to 42.78% of all cells in objects.csv. The most incomplete fields are:

Table / field	Missing count	Missing (%)	Potential impact and suggested review
offices.csv / created_at	112,718	100.00	No usable timestamps in this field; exclude from features unless the source can restore it.
offices.csv / updated_at	112,718	100.00	Same; retaining an all-missing field adds no information.
objects.csv / closed_at	459,842	99.39	Likely absent for many non-closed records, but semantics must be confirmed; missingness may itself reflect lifecycle status.
objects.csv / short_description	455,034	98.35	Text coverage is very sparse; most records cannot use this field without an explicit missing-text strategy.
objects.csv / first_investment_at	445,695	96.34	Likely structural for non-investors; do not treat absence as a date or ordinary random missingness.
objects.csv / last_investment_at	445,695	96.34	Same; consider a separate indicator only when justified by task timing.
objects.csv / first_funding_at	431,144	93.19	May be structurally absent for unfunded entities; verify before imputing.
objects.csv / last_funding_at	431,144	93.19	Same; potential leakage if later than a prediction cutoff.
objects.csv / state_code	407,891	88.16	Location detail is incomplete; consider broader geography or an explicit unknown category if appropriate.
people.csv / birthplace	167,872	74.04	Sparse personal data; likely exclude unless clearly relevant and ethically appropriate.

Table / field	Missing count	Missing (%)	Potential impact and suggested review
ipos.csv / source_description	1,086	86.26	Source notes are largely absent and are not ordinarily a predictive measure.

The missingness chart also shows high missingness in objects.csv / parent_id (about 94%), people.csv / birthplace, IPO source fields, objects.csv / city, and acquisitions.csv / term_code. The exact count for parent_id is not displayed in the saved missingness table excerpt, so it is not stated here. The notebook did not impute or remove missing values. Structural absence (for example, no IPO for a company) should not automatically be replaced with a statistical average.

4.2 Duplicates and key checks

- **Exact duplicate rows:** 0 reported in each of the 11 tables.
- **Candidate key checks:** The notebook's inferred candidate keys had zero missing key values and zero duplicate key rows. The fields checked were acquisition_id, id in degrees/investments/milestones/objects/people, funding_round_id, fund_id, ipo_id, office_id, and relationship_id.
- These are reassuring checks, but the notebook notes that candidate keys were inferred from field names. They do not validate all cross-table foreign-key relationships or prove that a proposed join has the correct grain.

4.3 Types, categories, and low-variance fields

- Date-like fields are stored as strings in the source data. The notebook's sampled non-missing date values were 100% parseable in checked fields. This is a parseability check, not a type conversion; the source columns remain unchanged.
- In objects.csv / region, the notebook found **16 case/space-normalization collision groups** and zero rows with leading or trailing spaces. This suggests values that may become duplicates after case/spacing normalization; it does not establish that the labels are semantically identical. Review before standardizing.
- Constant or near-constant fields include milestones.csv / milestone_code (one value, other, across 39,456 rows), funding_rounds.csv / pre_money_currency_code (one observed value, with many missing records), and the all-missing office timestamps. pre_money_valuation and its USD counterpart have 27 unique values but a 99.94% most-common value. These fields can add little predictive information or encode missingness rather than useful variation.
- No domain dictionary was available to validate all category labels. Category validity and inconsistent codes therefore remain unresolved.
- IDs, URLs, long text, and data-entry/audit timestamps may be unsuitable as direct predictors. Their utility depends on the task and leakage review.

4.4 Detected versus fixed

The checks detected missingness, potential category-format collisions, low-variance fields, and numerical outlier flags. **The notebook did not fix these issues**: no records were deleted, no missing data were imputed, no category normalization was applied, and no outliers were removed or capped.

5. Exploratory Data Analysis

5.1 Descriptive statistics

The notebook excludes identifier fields from numerical summaries because numeric-looking IDs are keys rather than measurements. It reports count, mean, median, standard deviation, minimum, quartiles, maximum, skewness, and kurtosis for the 22 numeric features. Selected measures are below; the complete displayed numeric summary is in Appendix B.

Table / feature	Mean	Median	Q1–Q3	Minimum–maximum	Skewness	Interpretation
funding_rounds.csv / raised_amount_usd	7,946,092	1,600,000	246,330–6,700,000	0–3,835,050,000	50.22	Mean is much higher than median, consistent with a strong upper tail.
funding_rounds.csv / participants	1.53	1	0–2	0–36	2.95	Most rounds have few participants; high counts are unusual under the quartiles.
acquisitions.csv / price_amount	388,619,100	0	0–500,000	0–2,600,000,000,000	97.61	Extreme values dominate the mean; zero values need domain interpretation.
funds.csv / raised_amount	399,481,300	100,000,000	16,300,000–275,000,000	0–89,000,000,000	30.16	Fund amounts are highly right-skewed with a wide range.

Table / feature	Mean	Median	Q1–Q3	Minimum–maximum	Skewness	Interpretation
objects.csv / funding_total_usd	892,672	0	0–0	0–5,700,000,000	147.41	At least 75% of values are zero under the reported quartiles; verify whether zeros mean no funding or unavailable data.
objects.csv / investment_rounds	0.174	0	0–0	0–529	73.99	Many zero counts and a small tail of very large values.
objects.csv / relationships	1.79	1	0–2	0–1,189	84.71	Most entities have few relationships in this field, with extreme maximums.
offices.csv / latitude	10.90	0	0–30.84	−43.77–69.65	1.05	The zero median and quartile warrant checking for placeholder coordinates.
offices.csv / longitude	−16.33	0	0–0	−159.48–176.92	−0.88	The zero quartiles warrant checking for placeholder coordinates; not proof that zero is invalid.

For each summary, **Q1** is the 25th percentile and **Q3** the 75th percentile; half of the recorded values lie between them. The standard deviation measures spread around the mean and can be inflated by extreme values. Skewness near zero indicates approximate symmetry; large positive skewness indicates a long right tail. Several financial fields have very high kurtosis, meaning unusually heavy tails relative to a normal distribution. Many reported zeros may be true zeros, placeholders, or missingness conventions; the notebook does not resolve which.

5.2 Categorical frequencies and company status

Company status was counted for **196,553 company records**:

Recorded status	Count	Percentage
Operating	183,441	93.33%
Acquired	9,394	4.78%
Closed	2,584	1.31%
IPO	1,134	0.58%
Total	**196,553**	**100.00%**

The breakdown is highly imbalanced: operating records outnumber each other class. If status were approved as a classification target, a model could achieve a misleadingly high accuracy simply by favoring the majority class. Class-aware metrics and stratified evaluation would be important. Status is not automatically a measure of success; operating could reflect the dataset's snapshot and recording rules.

For the top observed company categories, the notebook reports status percentages within each category. Examples: biotech records show 10.27% acquired, 2.64% closed, and 5.58% IPO; consulting shows 2.72% acquired, 0.62% closed, and 0.42% IPO; web shows 6.42% acquired, 4.04% closed, and 0.31% IPO. These are descriptive percentages for the selected categories, not adjusted risk estimates. Category coverage and missing category codes may affect comparisons.

6. Visualization Analysis

The figures below are existing notebook charts embedded from saved outputs; they were not recreated for this report.

Missingness

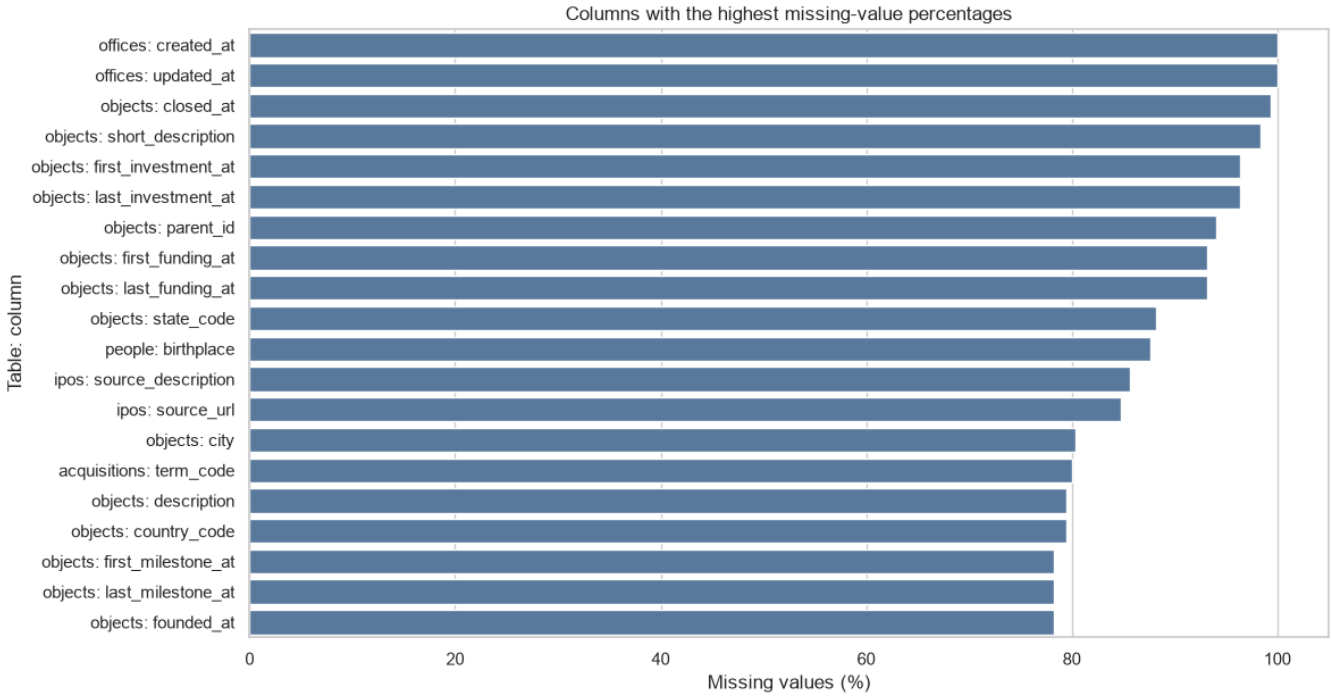


Figure 1. "Columns with the highest missing-value percentages." This bar chart ranks the 20 most incomplete fields. It is useful for prioritizing review because the highest bars are two fully empty office audit timestamps, followed by extremely incomplete company lifecycle and descriptive fields. The chart displays percentages, not the reason for missingness; it cannot distinguish "not applicable" from "not collected."

Candidate label distribution

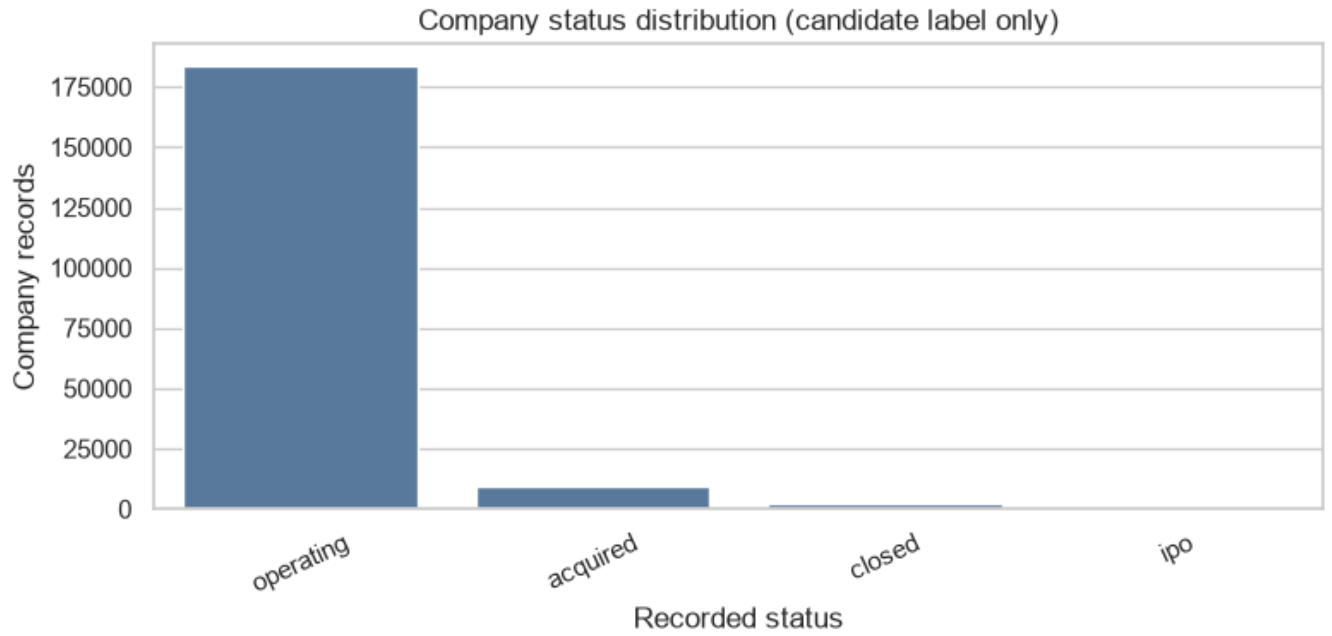


Figure 2. "Company status distribution (candidate label only)." The bar heights make the majority-class imbalance visible: operating records dominate the three less frequent outcomes. The plot is a count chart, so it does not adjust for time, sector, or company attributes and does not validate status as a target.

Numerical associations

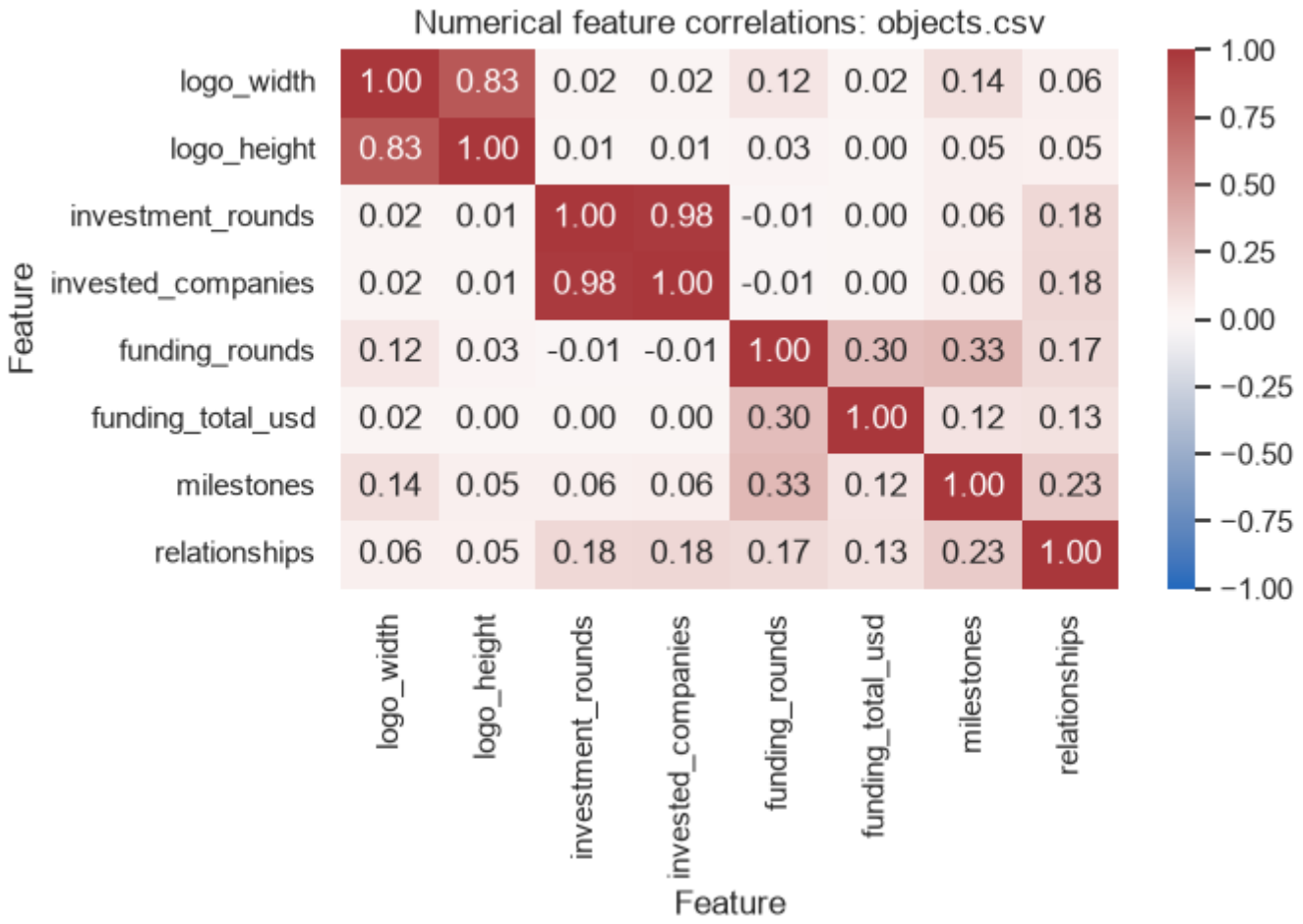


Figure 3. "Numerical feature correlations: objects.csv." This heatmap shows pairwise Pearson correlations among numeric fields in objects.csv. The strongest visible association is investment_rounds with invested_companies (r = 0.98); funding_rounds with funding_total_usd is weaker (r = 0.30). The heatmap summarizes linear association, not causation, and can be affected by zero-heavy distributions and snapshot aggregation.

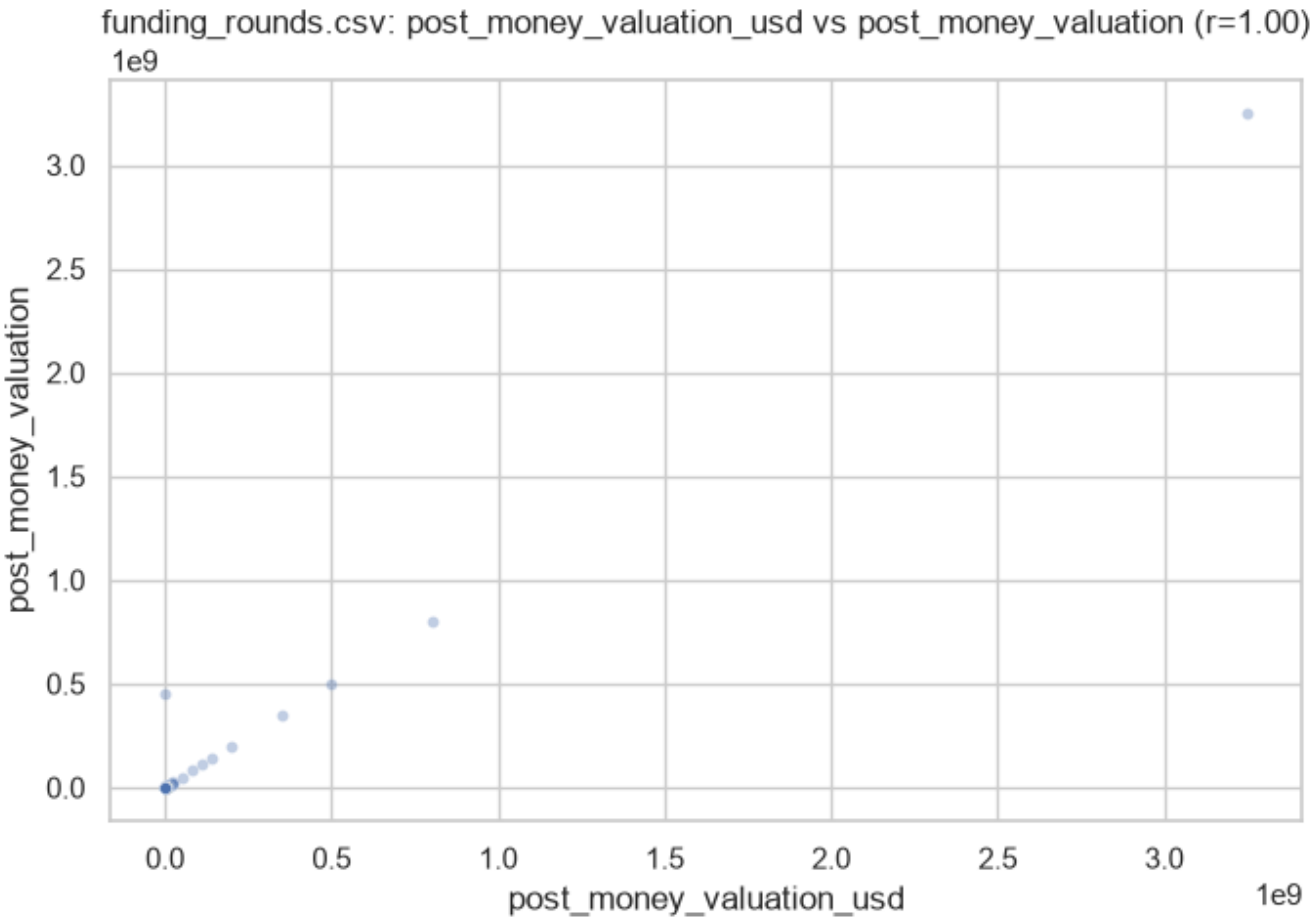


Figure 4. “funding_rounds.csv: post_money_valuation_usd vs post_money_valuation (r=1.00).” The scatterplot compares the reported valuation and its USD counterpart. The near-perfect line indicates the two columns are almost duplicates in this table; verify currency and conversion semantics before retaining both as features. Extreme values compress the central cloud, so the plot does not show fine-grained variation among smaller rounds.

Outlier visual

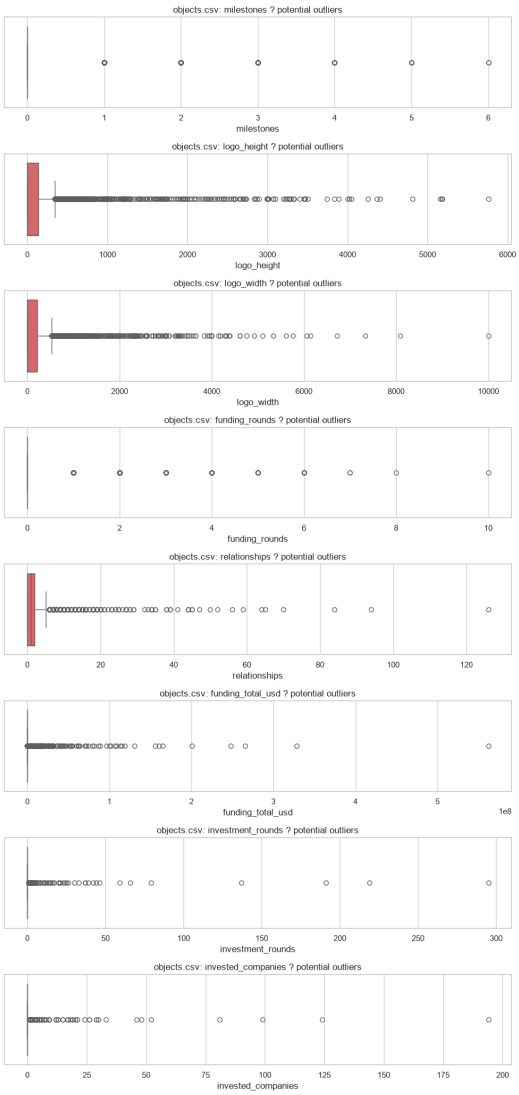


Figure 5. “objects.csv: potential outliers.” These boxplots show candidate IQR outliers for selected object-table counts and logo dimensions. Several count features have many zero observations, so the IQR rule's upper bound can be zero and label every positive count as an outlier. The plots flag unusual values, not errors; the logo dimensions may also include image-format or metadata artifacts.

The notebook additionally generated per-table numerical histograms/boxplots, low-cardinality categorical frequency plots, correlation heatmaps, and scatterplots for strongest numeric pairs. Large-table scatterplots and distribution plots use a seeded sample of up to 5,000 or 10,000 values, respectively, for legibility. No causal conclusions follow from any plot.

7. Outlier Analysis

The notebook applied two screening rules to the 22 numeric features:

- **IQR rule:** values below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR are flagged.
- **Z-score rule:** values more than 3 population standard deviations from the mean are flagged.

The bounds and counts below are calculated from non-missing observations. The IQR and Z-score counts need not match because they use different definitions. A large IQR count can occur for discrete or zero-inflated fields: for example, when Q1 = Q3 = 0, every positive value exceeds the IQR upper bound of zero. This does not mean every flagged observation is a data error.

Table / feature	IQR bounds	Observations	IQR flags (%)	Z > 3 flags (%)	Maximum
objects.csv / milestones	0 to 0	462,651	100,358 (21.69%)	16,769 (3.62%)	9
objects.csv / logo_height	−208.5 to 347.5	462,651	42,484 (9.18%)	7,636 (1.65%)	9,600

Table / feature	IQR bounds	Observations	IQR flags (%)	Z > 3 flags (%)	Maximum
objects.csv / logo_width	−324 to 540	462,651	38,436 (8.31%)	9,175 (1.98%)	18,200
relationships.csv / sequence	−2 to 6	402,878	37,654 (9.35%)	4,709 (1.17%)	399,946
offices.csv / longitude	0 to 0	112,718	32,368 (28.72%)	910 (0.81%)	176.92
objects.csv / funding_rounds	0 to 0	462,651	31,707 (6.85%)	11,173 (2.42%)	15
objects.csv / relationships	−3 to 5	462,651	30,199 (6.53%)	2,568 (0.56%)	1,189
objects.csv / funding_total_usd	0 to 0	462,651	27,874 (6.02%)	1,629 (0.35%)	5,700,000,000
objects.csv / investment_rounds	0 to 0	462,651	16,986 (3.67%)	1,208 (0.26%)	529
objects.csv / invested_companies	0 to 0	462,651	16,986 (3.67%)	1,490 (0.32%)	459
funding_rounds.csv / raised_amount_usd	−9,434,175 to 16,380,500	52,928	5,400 (10.20%)	287 (0.54%)	3,835,050,000
funding_rounds.csv / raised_amount	−9,275,000 to 16,125,000	52,928	5,395 (10.19%)	280 (0.53%)	3,835,050,000
funding_rounds.csv / participants	−3 to 5	52,928	2,487 (4.70%)	981 (1.85%)	36
acquisitions.csv / price_amount	−750,000 to 1,250,000	9,562	2,345 (24.52%)	1 (0.01%)	2,600,000,000,000
funding_rounds.csv / post_money_valuation	0 to 0	52,928	1,279 (2.42%)	34 (0.06%)	24,324,230,000
funding_rounds.csv / post_money_valuation_usd	0 to 0	52,928	1,194 (2.26%)	31 (0.06%)	24,324,230,000
funds.csv / raised_amount	−371,750,000 to 663,050,000	1,564	152 (9.72%)	10 (0.64%)	89,000,000,000
ipos.csv / raised_amount	0 to 0	1,259	138 (10.96%)	2 (0.16%)	24,400,000,000
ipos.csv / valuation_amount	0 to 0	1,259	108 (8.58%)	4 (0.32%)	108,960,000,000
funding_rounds.csv / pre_money_valuation_usd	0 to 0	52,928	31 (0.06%)	6 (0.01%)	15,000,000,000
funding_rounds.csv / pre_money_valuation	0 to 0	52,928	31 (0.06%)	6 (0.01%)	15,000,000,000
offices.csv / latitude	−46.2566 to 77.0943	112,718	0 (0.00%)	3 (0.003%)	69.6502

Notable range and distribution patterns:

- acquisitions.csv / price_amount has a maximum of 2.6 trillion and 24.52% IQR flags, but only one Z-score flag. This illustrates how a highly skewed distribution can cause the two rules to disagree. Validate units, currency, and extreme records against the source before interpreting the maximum.
- funding_rounds.csv / raised_amount_usd has 5,400 IQR flags but 287 Z-score flags. Extreme but valid funding rounds may be economically important; removing them could erase precisely the events the analysis aims to study.
- objects.csv / milestones, funding_total_usd, and several count measures have an IQR upper bound of zero because the middle half of values are zero. Positive counts are not automatically erroneous.
- offices.csv / longitude has 32,368 IQR flags due to Q1 = Q3 = 0; review coordinate completeness and possible zero placeholders, without assuming all flagged longitudes are invalid.
- Logo dimensions may reflect image assets rather than company characteristics and may be inappropriate predictors.

The notebook sampled up to five IQR-flagged row examples per feature for inspection. It did not verify them against an external source or apply any treatment. Recommended next step: confirm units and data semantics, inspect the most extreme valid records, then choose feature-specific handling (retain, log-transform, robust-scale, winsorize, or exclude) only if the project objective justifies it.

8. Feature Relationships

The notebook calculated correlations among non-identifier numeric fields within each table; tables were not automatically joined. The strongest reported pairs are:

Table	Feature pair	Pearson r	Interpretation
funding_rounds.csv	pre_money_valuation_usd and pre_money_valuation	1.0000	Near-duplicate representations; inspect currency semantics and avoid redundant features.
funding_rounds.csv	post_money_valuation_usd and post_money_valuation	0.9995	Nearly redundant; same caution.

Table	Feature pair	Pearson r	Interpretation
objects.csv	investment_rounds and invested_companies	0.9816	Very strong association between two investment-related counts; field definitions may overlap.
funding_rounds.csv	raised_amount_usd and raised_amount	0.9381	Strong association; likely related amount representations, but not necessarily identical.
objects.csv	logo_width and logo_height	0.8290	Larger image widths tend to accompany larger heights; likely an asset-size relationship, not a business outcome.
offices.csv	latitude and longitude	−0.6041	Geographic coordinates show a moderate negative linear association in the observed sample; this is not a geographic causal relationship.
ipos.csv	valuation_amount and raised_amount	0.3402	Moderate positive association in the recorded IPO fields.
objects.csv	funding_rounds and milestones	0.3304	Moderate association between event/summary counts.
objects.csv	funding_rounds and funding_total_usd	0.3049	Modest positive association between funding round count and total funding.

Correlation does not establish cause and effect. Near-duplicate amount fields could create multicollinearity or redundant information in a model. Some fields are summary counts of events recorded in the other tables, so a cross-table model must define a single entity-level row and a time cutoff. The notebook generated scatterplots for the strongest pairs, with samples used for legibility when tables were large.

The notebook also compared company status percentages across the top ten category_code groups. Biotech has the highest displayed IPO share (5.58%) among those categories; web has 4.04% closed; advertising has 5.12% acquired. These are raw within-category percentages and can reflect differences in category size, observation date, missing category coverage, or other confounders.

9. Machine Learning Readiness

No target was selected, no feature matrix was built, and no model was trained or evaluated. The notebook explicitly describes status as a candidate label only.

Readiness area	Current evidence	Work needed before training
Target and objective	Company status counts are available, but no objective or target is approved.	Define the business question, outcome classes or numeric target, and prediction horizon.
Unit of analysis	11 related tables have different grains.	Define one row per prediction unit; join or aggregate with validated keys and prevent duplicate entity leakage.
Missing values	High and likely structural missingness exists in some fields.	Investigate why values are missing; impute within training folds only; retain missingness indicators only when justified.
Dates	Date-like strings were sample-parseable but remain strings.	Convert and derive only date features known at the prediction cutoff.
Categories	Currency, status, geography, round type, and other categories are present.	Standardize labels with domain rules, then encode within a train-only pipeline; handle rare/unknown classes.
Scaling	Financial values have widely varying magnitudes and skew.	Consider log/robust transformations for scale-sensitive models after resolving zeros and units; tree models may need less scaling.
Class balance	Candidate company status is 93.33% operating.	If approved as target, use stratified splits, class-aware metrics, and possibly class weighting/resampling within training data.
Leakage	Status and later event/date/aggregate fields can reveal outcomes.	Exclude the target itself and features observed after the prediction time; time-bound funding, milestones, IPO/acquisition, and snapshot summaries.
Correlation/redundancy	Some numeric pairs have $r \approx 1$.	Review redundant currency representations and highly correlated features for the intended model.
Outliers	Many IQR flags occur in zero-heavy and heavy-tailed fields.	Verify extreme records and use domain-aware handling rather than automatic deletion.
Validation	The notebook contains no model validation.	Use temporal and/or company-group splits if the goal is future generalization; avoid the same entity appearing across train and test after joins.

If the intended task is company-status prediction, recorded status may be retrospective rather than a future label. A valid study would need a historical snapshot: features must be known before a fixed prediction date, while the target must describe an outcome after that date. Without this design, reported model performance could be inflated by leakage.

10. Key Findings and Insights

Data quality

• The dataset has 11 tables, 1,500,237 total rows, and 154 columns; missingness is uneven and reaches 42.78% of cells in objects.csv.

• The notebook found 68 columns with at least some missingness. Two office audit timestamp columns are entirely empty.

• No exact duplicate rows were found, and all 11 inferred candidate keys passed their missing/duplicate checks.

• Date-like fields were strings, although checked non-missing samples were fully parseable. Source fields were not converted.

• objects.csv / region has 16 normalization-collision groups. Category values generally lack an external domain dictionary for validation.

Statistical and distribution findings

• Amount and count fields are frequently right-skewed. raised_amount_usd in funding rounds has median $1.6 million, mean about $7.95 million, maximum $3.835 billion, and skewness 50.22.

• Company funding_total_usd has median and Q3 equal to zero, while its maximum is $5.7 billion. This makes the interpretation of zero values especially important.

• Company status among company records is heavily concentrated in operating (93.33%).

Outliers and relationships

• The IQR rule flags 100% of objects.csv / milestones because its upper bound is zero; this is a consequence of a zero-heavy distribution, not proof of invalid records.

• acquisitions.csv / price_amount reaches 2.6 trillion and has a 24.52% IQR flag rate, but only 0.01% Z-score flags.

• Near-perfect correlations exist between some likely parallel amount/valuation fields; the strongest association among object counts is investment_rounds vs. invested_companies ($r = 0.9816$).

• None of these associations establishes causation or guarantees that an extreme record is erroneous.

Potential ML opportunity

• A company-level status classification study is possible only as a proposed direction. The label is imbalanced and the prediction date is undefined, so leakage control and a target definition must come first.

• The relational tables could support funding, investor-network, or event-history analyses after their join keys and record grains are validated.

11. Limitations

• **Source and context:** The notebook does not identify the dataset publisher, collection date, observation window, sampling method, or formal variable definitions.

• **Representativeness:** Historical business records may overrepresent entities with public funding, prominent outcomes, or better reporting. This is a potential bias, not quantified by the EDA.

• **Missingness:** The analysis reports counts, not a full missingness mechanism study. It does not establish whether absence is random, structurally inapplicable, or a source omission.

• **Outlier rules:** IQR and Z-score thresholds are generic screening rules. They can over-flag discrete zero-heavy data, understate unusual values in skewed fields, and cannot validate source accuracy.

• **Correlation:** Pearson correlation detects linear pairwise association and is sensitive to extreme values and coding conventions. It cannot establish causation and may conceal nonlinear patterns.

• **Sampling in plots:** Some numerical charts use a deterministic sample of up to 10,000 observations and scatterplots up to 5,000. Those charts aid readability but are not a replacement for the full numerical summaries.

• **Cross-table interpretation:** No automatic joins were done, so relationships between event tables and entities are not established by the correlation analysis.

• **Modeling:** No target, preprocessing pipeline, model, or validation metrics exist in the notebook. No predictive performance claim can be made.

• **Unanswerable from this dataset alone:** Why a company succeeded or failed, the causal impact of funding, whether an unrecorded event did not occur, or how a future startup will perform.

12. Recommendations and Next Steps

Priority 1 — Essential data-quality and project-definition work

1. Obtain the source citation, collection/snapshot date, field definitions, currency conventions, and any official data dictionary.

2. Confirm the project question, prediction unit, target definition, and prediction time before selecting features.

3. Validate candidate primary and foreign keys and document the grain of every join. Test that joining event tables does not multiply company-level records unintentionally.

4. Investigate missingness in high-value fields, especially company funding/lifecycle fields and office coordinates. Separate “not applicable” from “not recorded.”

5. Confirm whether zero amounts, counts, and coordinates are true zeros or source-specific placeholders.

6. Review the 2.6-trillion acquisition-price maximum, high funding-round values, extreme relationship sequences, and unusual logo dimensions against source records.

7. Resolve the 16 region normalization-collision groups using domain-approved rules; do not normalize all categories blindly.

Priority 2 — Optional transformations

• Convert validated date strings into datetimes and derive elapsed-time or calendar features only as appropriate.

• Consider log or robust transformations for heavily skewed positive financial variables after determining how zero and negative values should be represented.

• Evaluate removal of constant/all-empty fields and redundant currency-converted features after verifying definitions.

• Retain valid extreme financial events when they are part of the business question; avoid indiscriminate trimming.

Priority 3 — Feature engineering opportunities

• Derive company age or time-from-founding features only using dates available at a defined prediction cutoff.

• Aggregate funding and relationship histories to a company-level snapshot, using only events before the prediction date.

• Create careful indicators for event presence or structurally missing attributes where the domain supports them.

• Consider geography, category, and network summaries only after checking coverage, privacy, and join consistency.

Priority 4 — ML preprocessing and validation

• Implement imputers, encoders, scalers, and transformations in a pipeline fitted only on training data.

• Use a chronological split and/or group split by company where future or unseen-entity generalization is the aim.

• If classifying status, report per-class precision/recall, macro-F1, confusion matrix, and balanced accuracy alongside accuracy.

• Check for leakage from status, IPO/acquisition/closure fields, future milestones, later funding events, and precomputed snapshot totals.

Priority 5 — Model training and evaluation

After the target and leakage-safe design are approved, establish a simple baseline, compare appropriate models, tune using training/validation data only, and evaluate once on a held-out test set. Report the split strategy, metrics, uncertainty, and limitations. These steps are recommendations; no model training has occurred.

13. Final Conclusion

The existing analysis establishes that this is a substantial, connected startup-investment dataset with useful company, financing, people, geography, and event information. The notebook successfully profiled table sizes, missingness, inferred key quality, descriptive statistics, categorical frequencies, numeric correlations, candidate status distribution, and potential outliers without altering the source rows.

Its strengths are the breadth of related records, the lack of detected exact duplicate rows, and the zero-duplicate results for the inferred candidate keys. Its weaknesses for immediate modeling are substantial and uneven missingness, unresolved zero semantics and extreme values, string-encoded dates, incomplete domain documentation, redundant fields, and an unselected target across tables of different grains.

The dataset is **suitable for further descriptive analysis and may support a carefully scoped ML project**, but it is **not ready for model training as-is**. The next step is not automatic cleaning; it is to confirm provenance and the project objective, define a leakage-safe prediction snapshot, validate relationships and value semantics, then build a documented preprocessing and evaluation pipeline.

Appendix A: Field Guide

The following meanings are inferred from column names and the notebook's table descriptions, not confirmed by a source dictionary. Names such as id, object_id, and *_object_id should be treated as keys; *_at fields are date/time-like strings; *_code fields are categorical codes; URL and description fields are provenance or text fields.

objects.csv **(40 fields)**

Field	Name-based description
id	Record identifier for the object row.
entity_type	Entity class, such as Company, Person, or Product.
entity_id	Identifier associated with the entity type.
parent_id	Possible parent/hierarchy reference.
name	Display name.
normalized_name	Normalized or canonical name.
permalink	URL-friendly record slug.
category_code	Industry/category code.
status	Recorded lifecycle status.
founded_at	Founding date.
closed_at	Closure date, where recorded.
domain	Website domain.
homepage_url	Homepage address.
twitter_username	Social-media username.
logo_url	Logo asset location.
logo_width	Logo image width metadata.
logo_height	Logo image height metadata.
short_description	Brief text summary.
description	Longer text description.
overview	Overview text.
tag_list	Tags or keywords.
country_code	Country code.
state_code	State/province code.
city	City name.
region	Region label.
first_investment_at	First recorded investment date.
last_investment_at	Last recorded investment date.
investment_rounds	Investment-round summary count.
invested_companies	Invested-company summary count.
first_funding_at	First recorded funding date.
last_funding_at	Last recorded funding date.
funding_rounds	Funding-round count.
funding_total_usd	Recorded total funding in USD.
first_milestone_at	First recorded milestone date.
last_milestone_at	Last recorded milestone date.
milestones	Milestone count.
relationships	Relationship count.
created_by	Record creator/source user label.
created_at	Record creation timestamp.
updated_at	Record update timestamp.

Supporting table fields

Table	Fields and name-based description
acquisitions.csv	id, acquisition_id (record/event IDs); acquiring_object_id, acquired_object_id (entity links); term_code (deal-term category); price_amount, price_currency_code (reported price and currency); acquired_at (event date); source_url, source_description (provenance); created_at, updated_at (audit timestamps).
degrees.csv	id (record ID); object_id (entity link); degree_type, subject, institution (education attributes); graduated_at (date); created_at, updated_at (audit timestamps).
funding_rounds.csv	id, funding_round_id (record/event IDs); object_id (entity link); funded_at (date); funding_round_type, funding_round_code (round categories); raised_amount_usd, raised_amount, raised_currency_code (amount and currency representations); pre_money_valuation_usd, pre_money_valuation, pre_money_currency_code (pre-money valuation); post_money_valuation_usd, post_money_valuation, post_money_currency_code (post-money valuation); participants (participant count); is_first_round, is_last_round (round indicators); source_url, source_description, created_by (provenance); created_at, updated_at (audit timestamps).
funds.csv	id, fund_id (record IDs); object_id (entity link); name (fund name); funded_at (date); raised_amount, raised_currency_code (amount and currency); source_url, source_description (provenance); created_at, updated_at (audit timestamps).
investments.csv	id (record ID); funding_round_id (round link); funded_object_id (recipient link); investor_object_id (investor link); created_at, updated_at (audit timestamps).
ipos.csv	id, ipo_id (record/event IDs); object_id (entity link); valuation_amount, valuation_currency_code (valuation); raised_amount, raised_currency_code (amount raised); public_at (listing date); stock_symbol (ticker); source_url, source_description (provenance); created_at, updated_at (audit timestamps).
milestones.csv	id (record ID); object_id (entity link); milestone_at (date); milestone_code (milestone category); description (text); source_url, source_description (provenance); created_at, updated_at (audit timestamps).
offices.csv	id, office_id (record IDs); object_id (entity link); description, region, address1, address2, city, zip_code, state_code, country_code (location attributes); latitude, longitude (coordinates); created_at, updated_at (audit timestamps).
people.csv	id (record ID); object_id (entity link); first_name, last_name (name); birthplace (place); affiliation_name (affiliation).
relationships.csv	id, relationship_id (record IDs); person_object_id, relationship_object_id (person/entity links); start_at, end_at (relationship dates); is_past (past/current indicator); sequence (ordering/count-like field); title (role/title); created_at, updated_at (audit timestamps).

Appendix B: Numerical Summary

The notebook reports these statistics for every numeric feature it selected. std is sample standard deviation. All values are as displayed or rounded to four decimals in the existing notebook output.

Table / feature	Count	Mean	Median	Std	Min	Q1	Q3	Max	Skew
acquisitions.csv / price_amount	9,562	388,619,100	0	26,603,740,000	0	0	500,000	2,600,000,000,000	97.6088
funding_rounds.csv / raised_amount_usd	52,928	7,946,092	1,600,000	42,168,200	0	246,330	6,700,000	3,835,050,000	50.2157
funding_rounds.csv / raised_amount	52,928	8,056,120	1,565,056.5	44,799,140	0	250,000	6,600,000	3,835,050,000	48.1763
funding_rounds.csv / pre_money_valuation_usd	52,928	329,452.5	0	65,318,030	0	0	0	15,000,000,000	228.8399
funding_rounds.csv / pre_money_valuation	52,928	329,452.5	0	65,318,030	0	0	0	15,000,000,000	228.8399
funding_rounds.csv / post_money_valuation_usd	52,928	1,824,359	0	128,706,500	0	0	0	24,324,230,000	145.7047
funding_rounds.csv / post_money_valuation	52,928	1,862,279	0	128,768,600	0	0	0	24,324,230,000	145.4972
funding_rounds.csv / participants	52,928	1.5286	1	2.0602	0	0	2	36	2.9531
funds.csv / raised_amount	1,564	399,481,300	100,000,000	2,468,491,000	0	16,300,000	275,000,000	89,000,000,000	30.1553
ipos.csv / valuation_amount	1,259	360,780,000	0	5,173,185,000	0	0	0	108,960,000,000	19.5320

Table / feature	Count	Mean	Median	Std	Min	Q1	Q3	Max	Skew
ipos.csv / raised_amount	1,259	58,879,500	0	870,828,500	0	0	0	24,400,000,000	24.8616
objects.csv / logo_width	462,651	189.1598	0	432.4035	0	0	216	18,200	6.5283
objects.csv / logo_height	462,651	130.4887	0	337.1916	0	0	139	9,600	6.9366
objects.csv / investment_rounds	462,651	0.1739	0	3.6380	0	0	0	529	73.9858
objects.csv / invested_companies	462,651	0.1381	0	2.5492	0	0	0	459	76.2947
objects.csv / funding_rounds	462,651	0.1137	0	0.5242	0	0	0	15	7.2663
objects.csv / funding_total_usd	462,651	892,672.2	0	17,001,200	0	0	0	5,700,000,000	147.4090
objects.csv / milestones	462,651	0.2652	0	0.5686	0	0	0	9	2.7323
objects.csv / relationships	462,651	1.7943	1	5.4567	0	0	2	1,189	84.7086
offices.csv / latitude	112,718	10.9014	0	18.9779	−43.7676	0	30.8377	69.6502	1.0478
offices.csv / longitude	112,718	−16.3292	0	43.9792	−159.4803	0	0	176.9163	−0.8841
relationships.csv / sequence	402,878	3,020.999	1	28,343.06	1	1	3	399,946	10.2192

Appendix C: Complete Outlier Screening Results

IQR bounds and rates are copied from the saved notebook outlier summary. Z is the number of observations with absolute Z-score greater than 3. Values have not been treated.

Table / feature	Min	Max	IQR lower	IQR upper	IQR count	IQR (%)	Z count	Z (%)
objects.csv / milestones	0	9	0	0	100,358	21.6919	16,769	3.6245
objects.csv / logo_height	0	9,600	−208.5	347.5	42,484	9.1827	7,636	1.6505
objects.csv / logo_width	0	18,200	−324	540	38,436	8.3078	9,175	1.9831
relationships.csv / sequence	1	399,946	−2	6	37,654	9.3463	4,709	1.1688
offices.csv / longitude	−159.4803	176.9163	0	0	32,368	28.7159	910	0.8073
objects.csv / funding_rounds	0	15	0	0	31,707	6.8533	11,173	2.4150
objects.csv / relationships	0	1,189	−3	5	30,199	6.5274	2,568	0.5551
objects.csv / funding_total_usd	0	5,700,000,000	0	0	27,874	6.0248	1,629	0.3521
objects.csv / investment_rounds	0	529	0	0	16,986	3.6714	1,208	0.2611
objects.csv / invested_companies	0	459	0	0	16,986	3.6714	1,490	0.3221
funding_rounds.csv / raised_amount_usd	0	3,835,050,000	−9,434,175	16,380,500	5,400	10.2025	287	0.5422
funding_rounds.csv / raised_amount	0	3,835,050,000	−9,275,000	16,125,000	5,395	10.1931	280	0.5290
funding_rounds.csv / participants	0	36	−3	5	2,487	4.6988	981	1.8535
acquisitions.csv / price_amount	0	2,600,000,000,000	−750,000	1,250,000	2,345	24.5242	1	0.0105
funding_rounds.csv / post_money_valuation	0	24,324,230,000	0	0	1,279	2.4165	34	0.0642
funding_rounds.csv / post_money_valuation_usd	0	24,324,230,000	0	0	1,194	2.2559	31	0.0586

Table / feature	Min	Max	IQR lower	IQR upper	IQR count	IQR (%)	Z count	Z (%)
funds.csv / raised_amount	0	89,000,000,000	−371,750,000	663,050,000	152	9.7187	10	0.6394
ipos.csv / raised_amount	0	24,400,000,000	0	0	138	10.9611	2	0.1589
ipos.csv / valuation_amount	0	108,960,000,000	0	0	108	8.5782	4	0.3177
funding_rounds.csv / pre_money_valuation_usd	0	15,000,000,000	0	0	31	0.0586	6	0.0113
funding_rounds.csv / pre_money_valuation	0	15,000,000,000	0	0	31	0.0586	6	0.0113
offices.csv / latitude	−43.7676	69.6502	−46.2566	77.0943	0	0.0000	3	0.0027